

Artistverse

Project Proposal

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Chapter 1 | Introduction and Rationale

In the current digital age, music has become a part of everyday life and a universal language that everyone can understand. Regardless of language, race, or culture, music helps listeners relax, release emotion, and relieve stress while at work or in their free time. Music is like a medium that helps communicate the feelings, emotions, and words of the composer or musician to the listener.

In composing music, the problem that most songwriters and musicians face is that it takes a long time to write the lyrics, rhythm, and key of a song. Including the chords to be used. Sometimes they just don't have enough ideas and don't know how to start writing lyrics. This is why we developed a website called Artistverse to solve the above-mentioned problems. Artistverse offers many features that make the songwriting process easier for users, songwriters, and musicians. Users can create lyrics based on their own ideas and preferences. Select the desired language for the lyrics, and even imitate the style of favorite artists. Including famous international artists like Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, and Thai artists like Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, and Da Endorphine. Users can also customize lyrics to reflect a variety of moods, including happy, sad, romantic, inspirational, energetic, chill, hopeful, reflective, mysterious, dramatic, nostalgic, playful, confident, dreamy, and empowering. Users can also choose the genre of their lyrics, including pop rock, country R&B, hip-hop, folk, indie, electronic jazz, blues reggae, classical, alternative, gospel, and punk.

Additionally, the chord progression feature also provides users with information about the keys on the guitar and piano and recommends appropriate chords based on the user's chosen key and progression. To ensure consistency and structure in song creation, Artistverse also offers innovative features such as a facial recognition camera system that analyzes users' moods and recommends songs based on their emotional state. Additionally, the platform can instantly analyze the lyrics about chords, rhythms, time signatures, and keys. It guides them through the creative songwriting process.

In summary, Artistverse is a platform that facilitates songwriting and creates an engaging experience for users. It helps make the songwriting experience easier and more enjoyable. Enabling users to efficiently create unique and interesting songs, Artistverse will be an essential part of music creation in today's and future digital eras.

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Chapter 2 | Literature Review

2.1 Business Review

2.1.1 Overview



Figure 1 songwriter

In today's era, music is not only a part of daily life. It is also an important tool for building relationships, relaxing, and stimulating our creativity. Today's technology increases connectivity. and creates a more perfect music experience. It provides online platforms and applications that help people create and access music more easily and conveniently. Songwriting is an important process in the music industry. It is not only about creating music for entertainment purposes. But it is also an important way for songwriters and musicians to communicate their feelings, thoughts, and experiences.

In today's world of online songwriting websites, there are many websites that offer similar services, but Artistverse is one such service that focuses on creating a unique and effective user experience. Meanwhile, Kukkik AI, HIX.AI, These Lyric Do Not Exist, WORDHERO and Rytr are websites that offer similar services. The emphasis is placed on creating lyrics with today's technology.

Artistverse helps songwriters and musicians efficiently create unique and interesting songs. The main features of the website include creating lyrics based on user ideas. Selecting the desired language in the lyrics Writing lyrics in the style of a favorite artist, choosing different moods in the lyrics, and selecting the desired music genre There are also chord suggestions based on the user's key and progression. Including other special features is a camera system that detects the user's facial expression, recommends songs that match the mood at that moment This feature utilizes facial expression analysis to recommend Spotify songs matching users' moods, enhancing their emotional connection to music. It's akin to having a friend who understands their emotional needs, elevating their listening experience. Additionally, it facilitates users in discovering musical references for creating their own songs based on emotions., and analyzes the user's lyrics about chords, tempo, time signature, and keys. With all its comprehensive features, Artistverse is a great way to create interesting and unique music in the digital age.

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2.1.2 Target

2.1.2.1 Problems

- Many online songwriting websites offer similar services. This makes it difficult for users to choose the right platform.
- Musicians and songwriters often have a difficult time composing new lyrics and finding inspiration for their lyrics.
- Beginning songwriters may lack the skills necessary to compose effectively.
- Music lovers may have trouble finding new music that matches their current mood or preferences.
- Musicians and songwriters may have difficulty creating chords for their songs due to a lack of expertise.

2.1.2.2 Solutions

- Users can use Artistverse tools to create lyrics from their own ideas and experiences. They can choose the desired language, select the music genre or mood of the lyrics to create, and easily select the names of famous artists to create lyrics in the desired style.
- The camera system detects facial expression and analyzes the user's mood, recommending songs that are appropriate for their mood at that time.
- The chord progression feature allows users to adjust the key and progression to achieve the correct chords according to common chord principles. That will make these chord sets not off-keys when played. With information about the keys of both guitar and piano.
- Artistverse gives users instant access to chords, rhythms, time signatures, and keys. This feature ensures that users will receive guidance from the lyrics and find more creative ways to inspire them.

2.1.3 Benefit

- Artistverse offers an easy-to-use platform designed to meet the needs of songwriters and musicians.
- Offering features including mood-based song recommendations and personalized lyrics creation, Artistverse makes it easier for users to improve their songwriting process and find inspiration.
- Artistverse gives users the flexibility to customize various aspects of their lyrics. Including music genres, moods, language, and writing lyrics according to the artist's style. This makes the lyrics more creative and unique.
- With the chord progression feature, songs can be diverse and not boring. by changing the instrument's sound over different chord progressions and creating an orderly and stable lyric structure.
- With instant access to chords, rhythms, time signatures, and keys. Users can improve their songwriting process. This feature provides quick advice and support provided by our real-time AI analysis.

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2.2 Business Tools / Software Review

There are some websites and applications similar to ours that aim to serve similar customers; for example, Kukkik AI, HIX.AI, These Lyric Do Not Exist, WORDHERO, and Rytr both have similar functions to our project but are different in many ways. The details are described below.

2.2.1 Kukkik AI

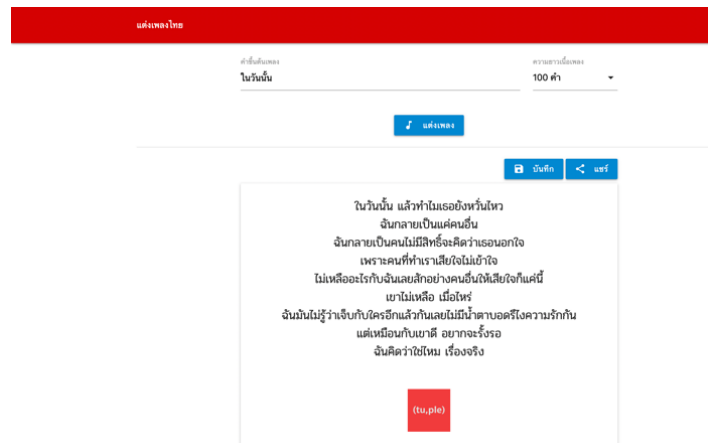


Figure 2 Kukkik AI

This website is a website that provides songwriting services. Users can choose how long they want their lyrics to be. There are 50–150 words to choose from, and they can input the beginning of the lyrics. [1]

Pros:

- Users can compose songs in both Thai and English.
- They have the option to choose the desired word count, allowing the song to be tailored to the desired length.
- Users can input the opening words for their songs.

Cons:

- Lyric length options are only 50, 100, and 150 words, which may not accommodate users who wish to compose longer songs.
- An accurate ultimatum sorting system results in mixing words and repeating phrases that are meaningless.

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2.2.2 HIX.AI

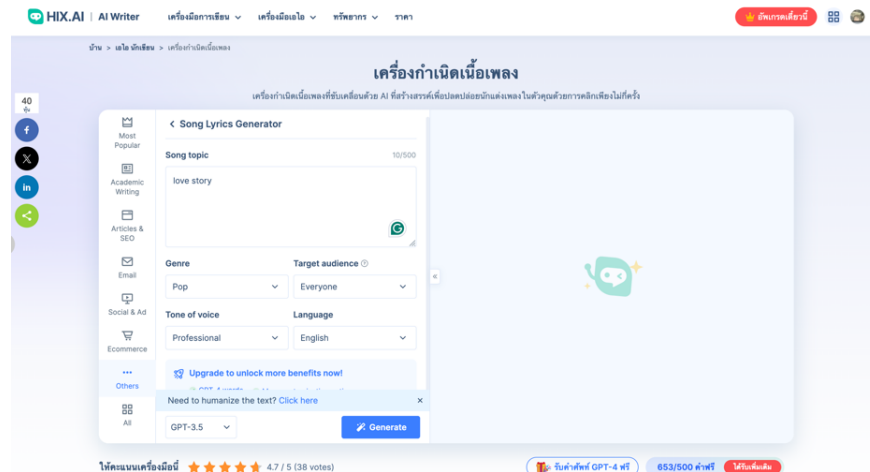


Figure 3 HIX.AI

This website is a website that provides songwriting services. Users can select the song topic for the lyrics they want. Choose what kind of genre they want. Who is the target audience for these lyrics Tone of voice, or even language. [2]

Pros:

- The website offers creative freedom and customization options for composing songs. Users can choose from various genres, target audience, and tone of voice for their lyrics.
- Users can choose from a variety of languages for song lyrics.
- The resulting lyrics are unique, rarely repetitive, and feature connecting words.
- It serves as an accessible platform for both budding songwriters and established artists seeking inspiration.

Cons:

- Limited free use with a small trial period.
- Subscription plans priced at \$7.99 USD/month, \$11.99 USD/month, and \$39.99 USD/month may be considered expensive by occasional users.

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2.2.3 These Lyric Do Not Exist

Generate Your Own Lyrics

Type a song topic word and then press the "Generate My Lyrics" button
example Song Topics: love, party, dance, happy, or human names such as peter, jennifer, bella, alex.
or you can even generate songs by emoji 🍷👯💃🥳🎉🤩🔥🌺🐼🦊🐶🐱🐭 or etc...

Song Topic:

Lyrics Genre:

 Country	 Metal	 Rock	 Pop	 Rap	 EDM
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Lyric Mood:

 Very Sad	 Sad	 Neutral	 Happy	 Very Happy
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Generate My Lyrics

Figure 4 These Lyric Do Not Exist

This website is a service that allows songwriters to compose lyrics according to their style. Users can input the song topic, lyric genre, and lyric mood of the desired lyrics. [3]

Pros:

- Users can select the desired song topic, lyrics genre, and mood of the lyrics as desired to get lyrics that match their needs and feelings.

Cons:

- The options for composing songs are not very diverse, as they are limited to choosing the song topic, lyric genre, and lyric mood.

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2.2.4 WORDHERO

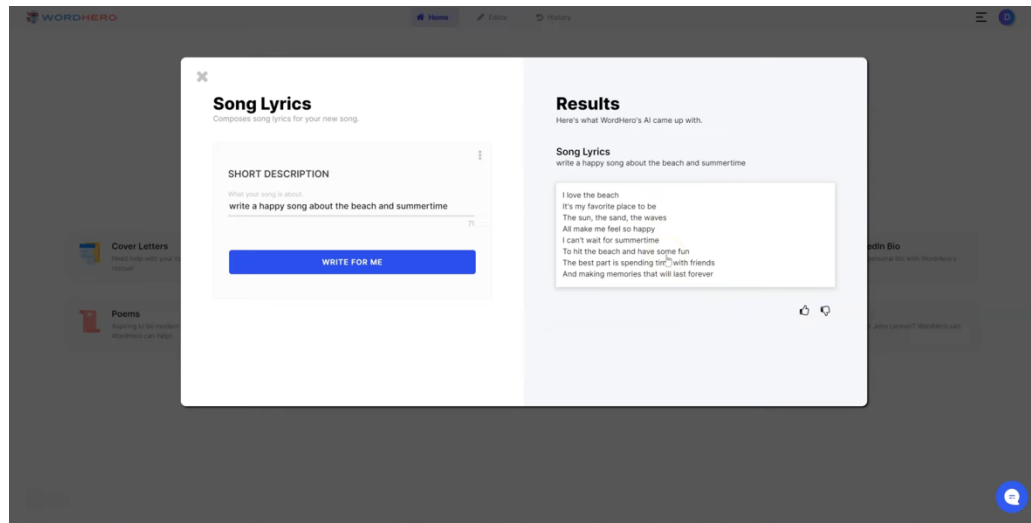


Figure 5 wordhero

This website is a service that allows songwriters to compose lyrics according to their style. Users can input the short description of their song lyrics. [4]

Pros:

- Users can input song ideas for their lyrics.

Cons:

- The options for composing songs are not diverse, and the lyrics that came out weren't very pretty. There is no connecting word in the lyrics, making the lyrics seem a bit disorganized.

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2.2.5 Rytr

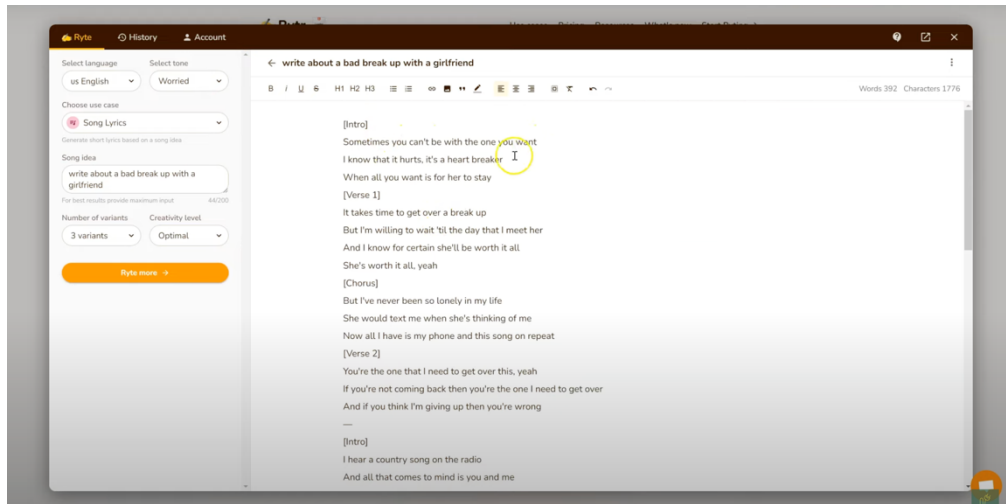


Figure 6 Rytr

This website is a service that allows songwriters to write lyrics in their own style. Users can enter song ideas for lyrics and choose the language they want. The tone of the lyrics can be selected. The user can select the number of variables and the creativity level of the lyrics. [5]

Pros:

- The user can compose lyrics by choosing the language, tone, and song idea of the lyrics as the user wants.
- The user can select the number of variants of the lyrics and adjust the creativity level of the lyrics, allowing the user to control the level of creativity desired.

Cons:

- The resulting lyrics do not have any connecting words. It makes the lyrics look disorganized.

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2.2.6 Feature Comparison

Features	Kukkik AI	HIX.AI	These Lyric Do Not Exist	WORDHERO	Rytr	Artistverse
Input the desired opening words or song topic for the lyrics	✓	✓	✓	✓	✓	✓
Choose the length of the lyrics	✓					
Choose language	✓	✓			✓	✓
Choose the genre of the lyrics		✓	✓			✓
Choose tone of voice, or target audience of the lyrics		✓				
Choose lyric mood			✓			✓
Choose a writing style for the lyrics according to the artist						✓
Recommend song by emotion						✓
Chord progression						✓
AI Lyrics Analysis						✓

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Chapter 3 | Development Tools Review



Figure 7 Visual Studio Code

Description

Visual Studio Code, or VSCode, is a code editor program used to edit and customize code. From the Microsoft camp It is developed in the form of open source, supports use on Windows, macOS, and Linux, and supports both JavaScript, TypeScript, and Node. js C++, C#, Java, Python, PHP, or Go Themes Debugger commands, etc. [6]

Alternative technologies

- Eclipse

Reasons for this choice

We will be using Visual Studio Code as the main code editing and customizing tool for frontend development using the Vue.js framework. VS code has syntax-highlighting capabilities that help make the code clear and easy to understand. Including the use of features Auto-completion That helps write code quickly and accurately. The program will help to complete the words automatically. There are also benefits to using extensions, for example, tailwind vue.js. Additionally, there is a debugger tool that helps to quickly detect and fix errors in the code, making the work process more efficient.

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Figure 8 Vue Framework

Description

Vue.js is a JavaScript framework used to build full-featured web applications. The user interface is beautiful and easy to use. Able to create websites using the languages used: HTML, CSS, and JavaScript. Vue.js can be used in various parts, including creating components. repeated use Application data management and status determination Manage API calls and events that occur in your application. [7]

Alternative technologies

- React

Reasons for this choice

We will use Vue.js as a tool for creating user interfaces that display results instantly in real time when information on the website changes. It can track data changes and update the interface automatically. Support for creating reusable components and modules.

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Figure 9 GitHub

Description

GitHub is a website that provides Git (Version Control Repository) services as well as a source code repository for many large projects, including Kubernetes, NET, Node.js, Python, PyTorch, and Swift, helping to facilitate the development of software and make the work process go more smoothly and quickly. [8]

Alternative technologies

- GitLab

Reasons for this choice

We will use GitHub as a collaboration tool for sharing code and developing it together. It helps in managing the code and tracking code changes. Track project progress, report issues, and collaborate on effective solutions.

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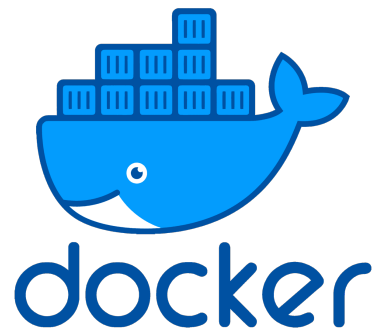


Figure 10 Docker

Description

Docker is an open-source tool or software platform. that helps to easily build, test, and deploy applications. Quick-to-run services or servers according to the principles of building containers to manage various libraries. It also helps manage version control to make it easier to deal with any problems that arise. [9]

Alternative technologies

- VM

Reasons for this choice

Docker makes deploying applications easier and more efficient. It can create Docker images with applications and environments prepared and ready to use. Which can be deployed on local servers, the cloud, or even on cluster management systems.

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Figure 11 PyCharm

Description

PyCharm is an IDE (Integrated Development Environment) developed by JetBrains that is specifically for programming in Python. It has many features that help developers work easily and efficiently, including intelligent coding with word completion, introducing and displaying help messages, managing version control, testing code, and working with the debugger camera. [10]

Alternative technologies

- Visual Studio code
- Jupyter Notebook/JupyterLab

Reasons for this choice

We will use PyCharm as a tool for writing Python and Flask to develop the backend, as it supports web development frameworks like Flask and its powerful and versatile tooling, and the interface is friendly and easy to use. Multiple platforms are supported, and PyCharm has extensive Python support. PyCharm's comprehensive features streamline our project development process, making it easier to efficiently write, test, and edit Python code.

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Figure 12 Flask

Description

Flask is a very popular Python web framework. With simplicity in use, learning to understand the structure is not complicated. It makes it possible to write freely. It is designed to be concise and emphasize simplicity. Lightweight and straightforward, Flask is flexible and allows developers to build custom web applications according to their project needs. It's good and facilitates systematic web application development. (microservices) with easy integration with other frameworks. [11]

Alternative technologies

- FastAPI

Reasons for this choice

We will use the Flask framework as a service for web servers to pass requests between the frontend and backend, facilitating smooth communication and data exchange across different layers of our web applications.

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Figure 13 Amazon EC2

Description

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides scalable computing space in the cloud. It's designed to help developers make web-scale computing easier, with users paying as they go. Without having to invest in building or maintaining the hardware, they can rent servers according to their needs. and the server can be scaled according to application or business needs. [12]

Alternative technologies

- Microsoft Azure Virtual Machines

Reasons for this choice

We will use Amazon Elastic Compute Cloud (EC2) to provide the flexibility to scale computing resources in the cloud. This allows us to freely adjust the size of our servers according to the needs of our websites and applications. It also costs per usage, eliminating the need for an upfront investment in hardware. This makes it convenient to have a system that responds to their needs appropriately.

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Figure 14 Jupyter Notebook

Description

Jupyter Notebook is an application used for writing and running programs in Python and other programming languages. It has the ability to combine code, text, images, and other media into interactive documents called notebooks. Code can be written and run interactively right away. This makes it the perfect tool for experimentation. Program development and displaying data efficiently. [13]

Alternative technologies

- PyCharm
- Visual Studio code

Reasons for this choice

We will use Jupyter Notebook as a tool to dump models to send to the flask server for functions. Jupyter helps in developing and testing machine learning models efficiently and testing the functionality of the code in a responsive format. Model results can be easily displayed using libraries like Matplotlib, Seaborn, and Plotly.

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Figure 15 MySQL

Description

MySQL is a database management system, or Relational Database Management System (DBMS) or Relational Database Management System (RDBMS), which is a database system that stores data in a table format. The data is divided into rows (rows), and each row is divided into columns (columns) to link the data in the table with the data in the specified column. Instead of keeping separate data without any connection. [14]

Alternative technologies

- MongoDB

Reasons for this choice

We will use MySQL as a data management tool. It can support accessing large amounts of data simultaneously without degrading performance.

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Chapter 4 | Quality Standard

ISO 29110 for Very Small Entity (VSE)

ISO/IEC 29110-4-1:2011 is applicable to Very Small Entities (VSEs). A Very Small Entity (VSE) is defined as an enterprise, organization, department, or project having up to 25 people. A set of standards and guides has been developed according to a set of VSEs' characteristics and needs. The guides are based on subsets of appropriate standards elements, referred to as VSE profiles. The purpose of a VSE profile is to define a subset of International Standards relevant to the VSE context.

4.1 Project Management Process

The purpose of the software management process is to establish, develop and carry out systematically the task of the software implementation project, which allows complying with the project's objective in the expected quality.

The 4 activities in the project management process

1. Project Planning
2. Project Plan Execution
3. Project Assessment and Control
4. Project Closure

4.2 Software implementation process

The purpose of the software implementation process is the systematic performance of the analysis, design, construction, integration, and test activities for new or modified software products according to the specified requirements. There are 6 Activities.

1. Software Implementation Initiation
2. Software Requirement Analysis
3. Software Architectural and Detailed Design
4. Software Construction
5. Software Integration and Tests
6. Software Deliver

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Chapter 5 Software Configuration Management

5.1 Naming Convention

The filename format that we using for all document is

Artistverse-[File Name] V.[Version].[FileType]

- File Name: The name of each process phase.
- Version: This part is the version of the file. The version number will be in this format “[Main version].[Subversion]”
- File Type - The type of the file format for example .docx or .pdf.

5.2 Change Management

Change management manages all of the changes in the project during the development process. All the change requests will be recorded into the change record document. The procedures for managing changes are:

1. Discuss with an advisor about the change.
2. Record the change information to change the document.
3. Send the change request to an advisor.
 - 3.1 Request accepted: change document and software follow the change information.
 - 3.2 Request rejected: continue and find alternative solutions.

5.3 Project Repository

- Microsoft Word: For document management
- GitHub: For source code management

Item	File Name	File Type	Baseline Version
Project Proposal	Artistverse-Proposal V.[Version]	.docx	2.0
Project Plan	Artistverse-ProjectPlan V.[Version]	.docx	2.0
Software Requirement Specification	Artistverse-SRS V.[Version]	.docx	2.0
Software Design Document	Artistverse-SDD V.[Version]	.docx	2.0
Test Plan	Artistverse-TestPlan V.[Version]	.docx	2.0
Test Record	Artistverse-TestRecord V.[Version]	.docx	2.0
Traceability Record	Artistverse-TraceabilityRecord V.[Version]	.docx	2.0

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Item	File Name	File Type	Baseline Version
Software Product	Artistverse-SoftwareProduct V.[Version]	.docx	2.0

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Chapter 6 Literature Review

6.1 Motivation

The development of Artistverse came from an understanding of the importance of music in people's daily lives. This is not just listening to music for entertainment purposes only. But it is also a way to communicate feelings and ideas. Music helps us relax, reflect on our emotions, and hone our creativity. We saw the need for a platform that empowers songwriters and musicians to create lyrics. But sometimes there are obstacles in the music creation process. Sometimes they have trouble writing lyrics and lack the creativity to begin with, so Artistverse was created to solve that problem. It provides tools and features that help users create music with ease and convenience in today's digital era.

6.2 Aims and Objective

6.2.1 Aims

Artistverse aims to help songwriters and musicians create unique and engaging music efficiently in today's digital age.

6.2.2 Objective

- A website allows and enhances the songwriting experience on the website by providing users with a comprehensive set of tools and features. This includes allowing users to compose songs based on their prompt lyrics topic and displaying the lyrics' content as they enter it. Additionally, the platform offers flexibility by enabling users to select the language and music genre for their lyrics. Moreover, users can choose the mood and artist style they wish to convey in their lyrics.
- A website that detects facial expression and recommends songs that match the user's mood in real time.
- A website that helps users know the chord progressions for the songs they want to create.
- A website ensures users receive prompt guidance and analyze the lyrics by AI Analysis instantly.

6.3 Deliverables and Limits

6.3.1 Deliverables

- Proposal
- Software project plan document
- Software requirement specification
- Software design document
- Software test plan document
- Software test record document
- Software traceability document
- Software product

6.3.2 Limits

1. Artistverse is a website that required an internet connection.
2. System support only English interface.
3. The system cannot produce a melody.

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4. Accuracy depending on gpt-3.5-turbo-1106 model .
5. Song that recommend in the feature mood detection come from playlists that categorized into mood-specific selections in Spotify.
6. The data of the model used to train facial expression capture correctness is limited to the Sequential Model from Keras, which trained by using only the facial of the Westerners

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6.4 System Architecture

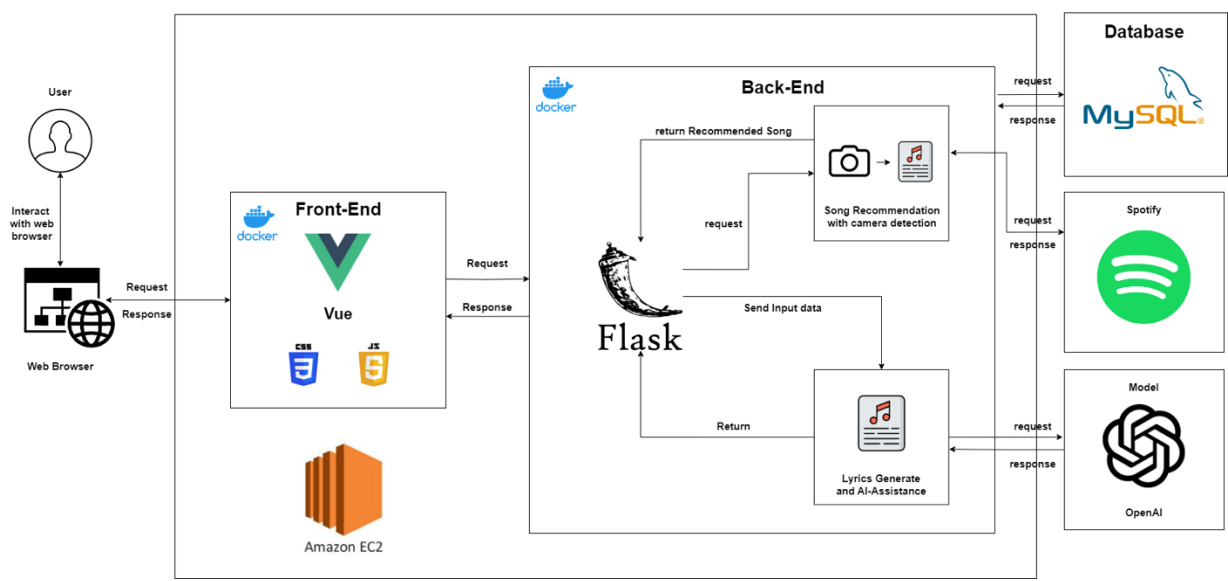


Figure 16 System Architecture

Description:

The system architecture consists of a front-end built using Vue.js, a lightweight Python web framework called Flask, and Docker for containerization. The front-end allows us to create interactive web applications using reactive data binding and components. The back end enables us to create web applications by defining routes and handling requests and responses. Docker allows us to package our application and its dependencies into a single, portable container, ensuring easy deployment and running on any platform that supports Docker. We used the OpenAI model for the function-generated lyrics and Ai-Assistance to get the return result of the lyrics generated and the question-answer based on the user input. Spotify Api is for getting a list of the songs to use in the function song recommendation. The MySQL database stores input, lyrics, and other relevant information, which is used to retrieve song lyrics and other data when generating recommendations.

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6.5 Software Process

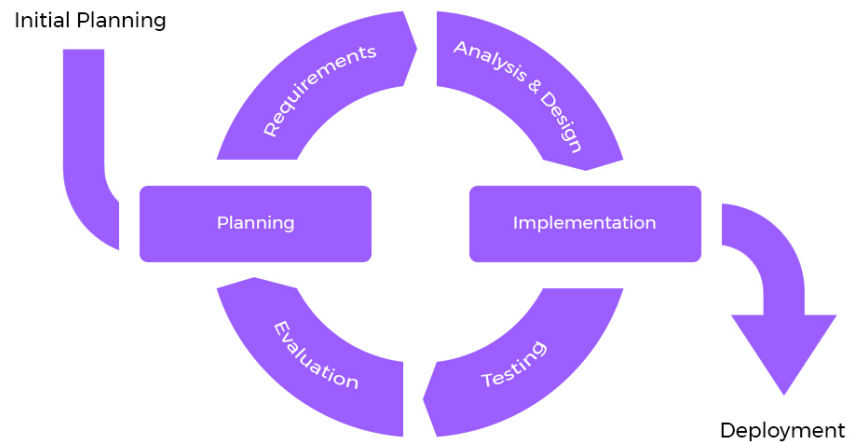


Figure 17 Iterative process

An iterative process model is a model used to develop software or software-related products. This model works in that the development process is divided into rounds, or iterations. Each round includes planning, design, development, and testing of the product. So that it can be improved and adjusted throughout development. It includes users or related parties in each round so that they can check and give suggestions for improving the product at all times. This model allows for flexibility in software development. Can be improved quickly and appropriately according to changes in needs and usage environments. [15]

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6.6 Feature

Feature #1: Dynamic Lyric Composition

Description: This feature empowers users to generate personalized song lyrics tailored to various criteria, including Mood, Genres, Languages, and Artist Styles. Users can input prompt words into our system and select mood, genres, languages, or even the style of their favorite artists, and the system will compose lyrics accordingly. Users can select from a range of mood tones, including happy, sad, romantic, inspirational, energetic, chill, hopeful, reflective, mysterious, dramatic, nostalgic, playful, confident, dreamy, and empowering. Additionally, users can specify the genre they desire, choosing from options that include pop rock, country R&B, hip-hop, folk, indie, electronic jazz, blues reggae, classical, alternative, gospel, and punk. Moreover, users can select the language in which they want the lyrics to be composed, with options ranging from English, Thai, Japanese and Chinese. Furthermore, users can compose lyrics in the style of their favorite artists, including international stars like Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, and Thai artists like Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, and Da Endorphine. By providing input prompts, users can initiate the lyric generation process, allowing the system to create unique and personalized lyrics that align with their preferences, emotions, language choices, genre preferences, or favorite artist styles. Users can save the generated lyrics to their computer in a text file.

Feature #2: Mood Detection Camera Integration

Description: This feature is all about understanding how much music influences our emotions. By picking songs from Spotify that match how users feel and using their facial expressions to know their mood, it helps users find music that really speaks to them emotionally. It's like having a friend who knows exactly what song they need to hear when they're feeling sad or happy, making the listening experience more personal and enjoyable. It also assists users in finding musical references that inspire them to create their own songs based on their emotions. Our website incorporates a mood detection camera feature that analyzes users' facial expressions to determine their current mood, including moods such as sad, angry, fearful, surprise, happy. The feature includes photo capture functionality to capture the user's facial expression based on the detected mood, and the system recommends curated songs in playlists from Spotify using the Spotify API. These songs in playlists are categorized into mood-specific selections, including sad, angry, surprised, fearful, and happy playlists. The recommended playlists are updated monthly to ensure fresh content. Users can interact with the recommended playlists by clicking on a song, which redirects them to the corresponding Spotify application to listen to the selected song track. Additionally, the system dynamically detects the top 3 songs from the user's selected song and displays them on our website for further exploration and enjoyment.

Feature #3: Chord progression

Description: This feature enables users to select the key and chord progression for their songs, utilizing fundamental chords in music theory. Users can input the desired key that has C,D,E,F,G,A,B, and select a chord progression sequence, such as the common progression that

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has 2-5-1-6,1-6-2-5,1-5-6-4,1-6-4-5,1-4-5-1,1-4-6-5,1-6-2-6,6-4-3-2. This is an example. If the user inputs key C and selects progression 1-4-5-1, the system will display the chord C-F-G-C. ("1-4-5-1" in the key of C means they want to use the first (C), fourth (F), fifth (G), and first (C) chords of the C major scale in that order) based on the user's. The system then displays the corresponding chords based on the selected key and progression. Additionally, to aid users in understanding and implementing the selected chords, the system provides visual representations of chord pictures for both piano and guitar. These pictures depict the finger positions for each chord on both instruments. Furthermore, users can hear the sound of each chord. This feature allows users to audibly confirm the chord's sound, ensuring accuracy and aiding in practice and composition.

Feature #4: AI Lyrics Analysis

Description: The AI Lyrics Analysis feature analyzes user-input lyrics to suggest chords, tempo, time signature, keys, and extracts musical information based on the provided lyrics, guiding users creatively. Users need to input lyrics only, and the system will provide a detailed musical analysis, including chord suggestions, tempo recommendations, time signature, and key identification based on their input lyrics.

6.7 User journey

A user journey is a concept used to describe the experience a user has throughout a website or application by understanding how users interact with a product or service from start to finish.

The journey begins when, upon entering the Artistverse website, users are presented with a homepage that showcases images, text, and videos related to the website. After that, in the top right bar of the website, when the user clicks, the system will display options including Homepage, Lyric Generator, Camera Mood, Chord progression, and AI-Analysis. After that, the user can select the desired option. After that, the system will take the user to that page. Figure 18 shows the homepage.

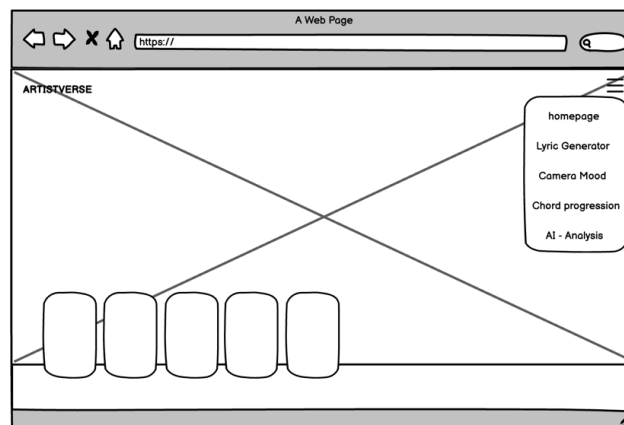


Figure 18 UI01-Homepage

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When the user clicks Lyric Generator, there are 3 options for the user to choose for composing lyric is Lyric Generator by Genre, Lyric Generator by Mood, and Lyric Generator by Style Artist, as shown in Figure 19.

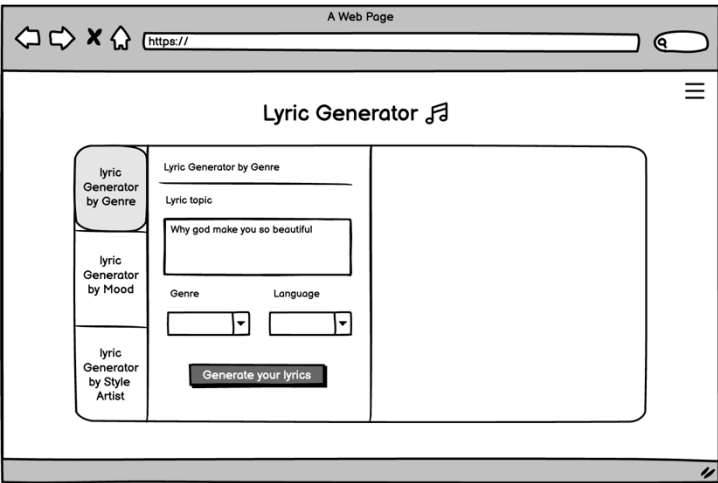


Figure 19 UI02-Lyric Generator

If the user input is only a lyric topic, the system will not be able to generate lyrics. The system will require the user to fill in all information completely before generating lyrics, as shown in Figure 20.

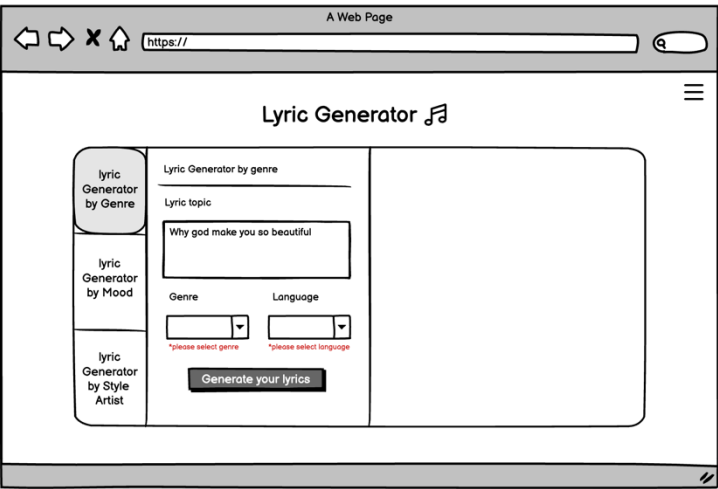


Figure 20 UI03-Lyric Generator with Lyric topic

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When the user clicks Lyric Generator by Genre, the user can input the lyric topic of their lyric and select the genre, choosing from options that include pop rock, country R&B, hip-hop, folk, indie, electronic jazz, blues reggae, classical, alternative, gospel, punk, and the language of the lyric, with options ranging from English, Thai, Japanese and Chinese. Moreover, the user can also save the lyrics, as shown in Figure 21.

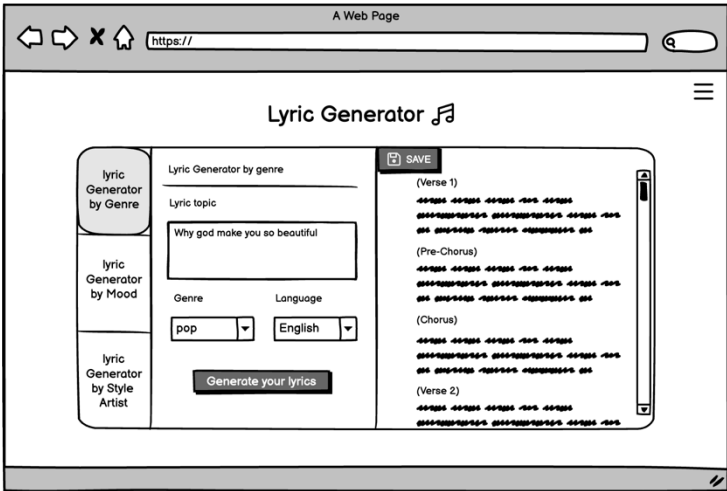


Figure 21 UI04-Lyric Generator by Genre

The user can choose the location to save the lyrics file. It will display a window for the user to choose where they want to save the file. The user gives the file a name and clicks the "Save" button to save the file to the selected location, as shown in Figure 22.

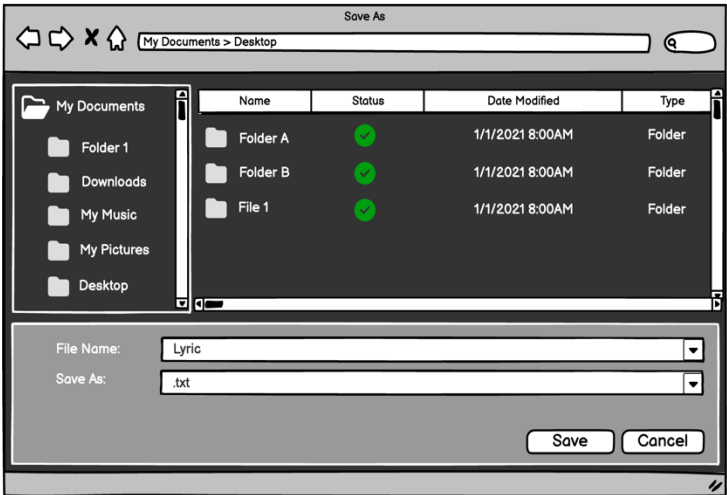


Figure 22 UI05-Save file

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When the user clicks Lyric Generator by Mood, they can input the lyric topic of their lyrics and select the mood, including happy, sad, romantic, inspirational, energetic, chill, hopeful, reflective, mysterious, dramatic, nostalgic, playful, confident, dreamy, and empowering. and language of the lyric they want, which are English, Thai, Japanese and Chinese. They can also save the lyrics, as shown in Figure 23.

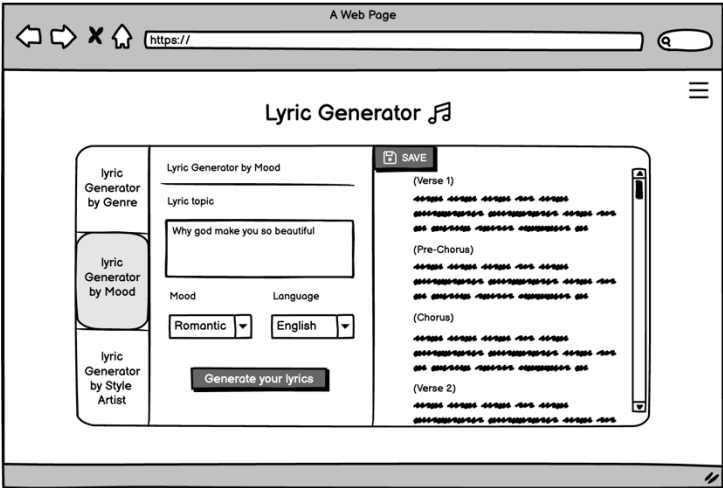


Figure 23 UI06-Lyric Generator by Mood

And when the user clicks Lyric Generator by Style Artist, they can input the lyric topic of their lyric and select the style artist, including international stars like Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, and Thai artists like Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, and Da Endorphine. and choose the language of the lyrics they want, which are English, Thai, Japanese and Chinese. Additionally, the user can save the desired lyrics, as shown in Figure 24.

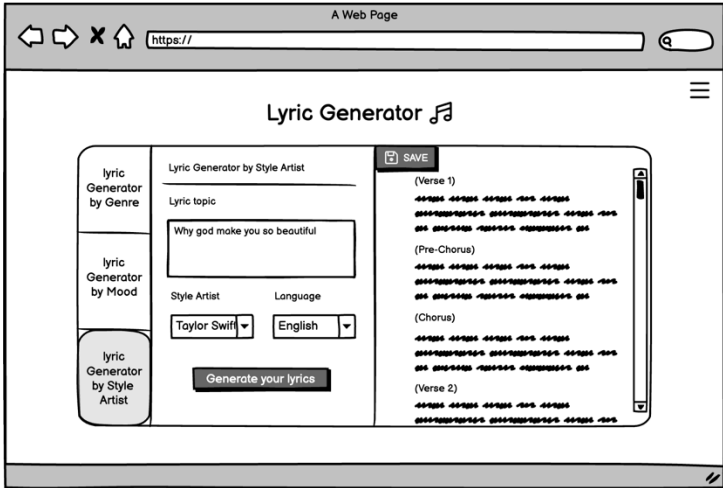


Figure 24 UI07-Lyric Generator by Style Aritst

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If the user clicks on the Camera Mood page, a message will be displayed asking the user whether to allow the camera to be turned on or not. If the user agrees, the system turns on the camera so the user can take a photo and detect their mood. Users can click the "capture" button to take a photo and detect their mood at that moment.

Once the system has finished detecting and identifying the user's emotions, it will display a playlist that suits the user's mood. The system will recommend the three most suitable songs for the user to choose from. When the user selects the song they want to listen to, they will be redirected to the Spotify application to listen to the selected song.

If the user wants the system to recommend new songs again, they can press the "capture" button again to take the photo and detect their mood again. The system will work as usual by recommending songs that suit the user's new mood, as shown in Figure 25.

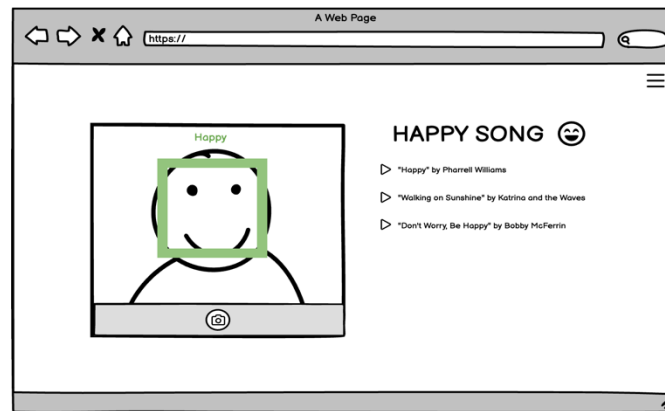


Figure 25 UI08-Camera Mood page

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When a user clicks on a chord progression, the website can provide additional information to help users who may not be familiar with the keys of guitar and piano to make user learn more information, and when users select the chord that they want, the system will provide the sound of the instrument corresponding to the selected key, as shown in Figures 26–27.

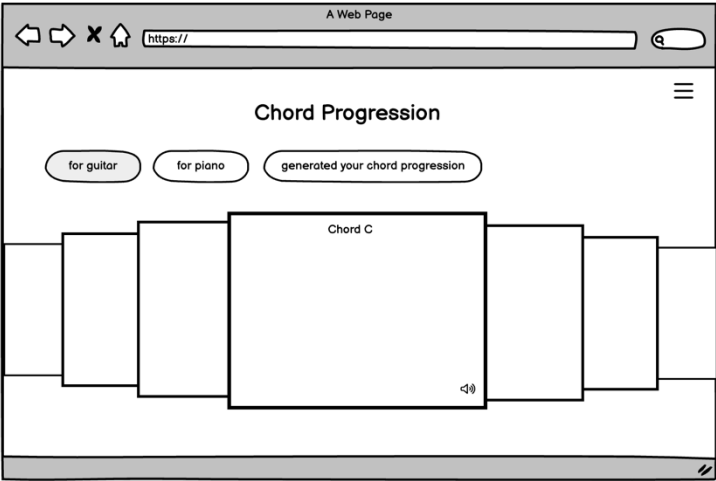


Figure 26 UI09-Chord Progression for Guitar

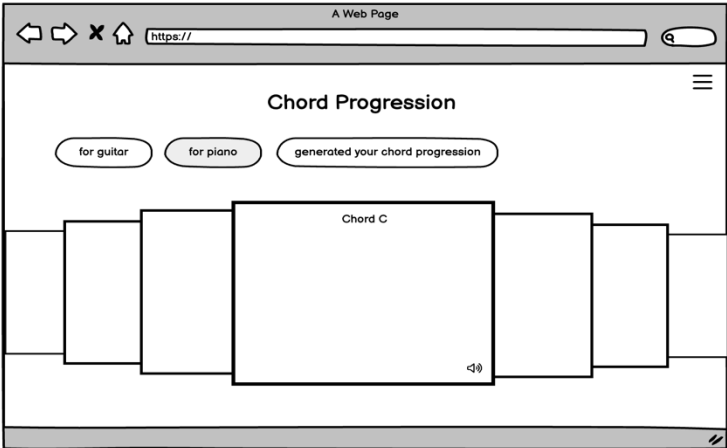


Figure 27 UI10-Chord Progression for Piano

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When the user wants to create a progression for a song, the user selects the key of the song and selects the desired progression. The system creates a progression based on the selected key and progression. The created progression is then displayed for the user to review and adjust as needed. If users want to hear how the progressions they created play out, they can press to listen, as shown in Figure 28.

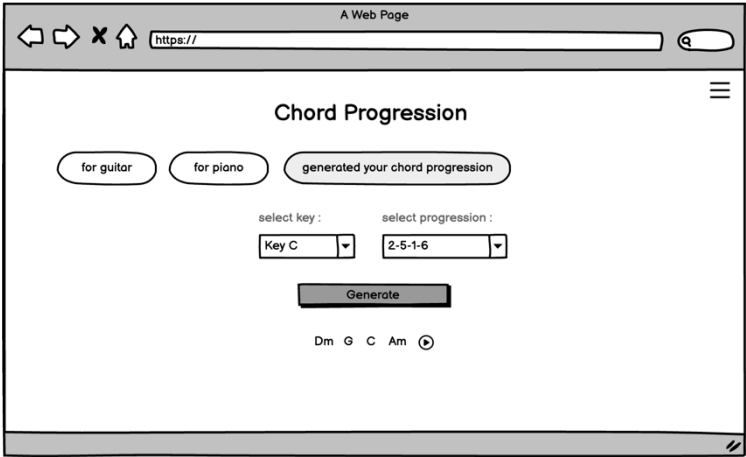


Figure 28 UIII-Generate Chord Progression

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When the user clicks on the AI-Analysis page, they are directed to a dedicated section where they can interact with the AI Analysis. The user can input the lyrics that they want to analyze into our system. The system will analyze user prompt lyrics and provide suggestions, including chords, tempo, time signature, keys, and extracts musical information based on the user prompt lyrics, as shown in Figure 29.

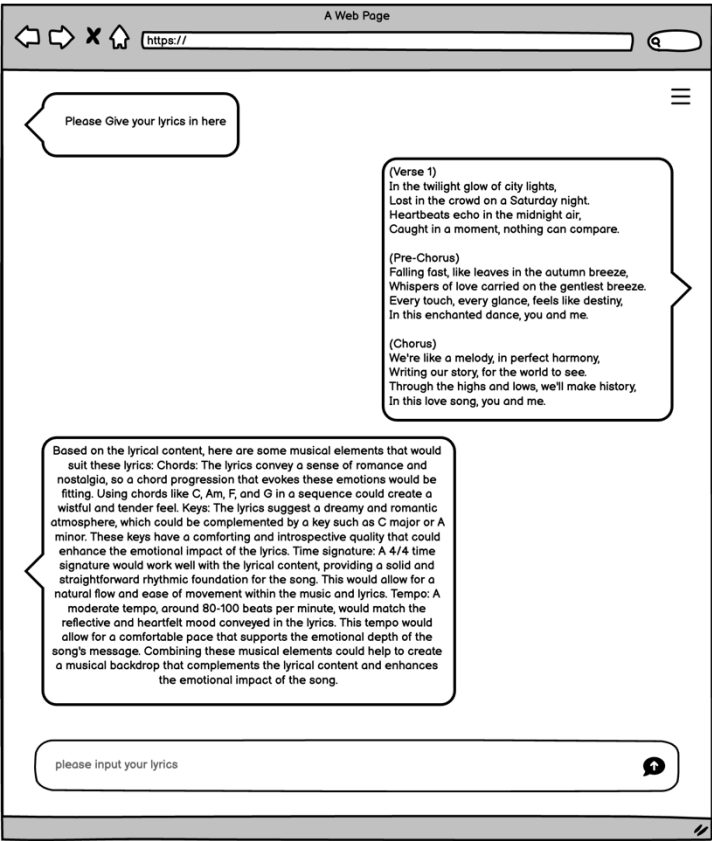


Figure 29 UI12-AI Analysis page

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6.8 Schedule Plan

6.8.1 Progress Report

Normal plan

Senior Project Event	Deliverables
Proposal Presentation	Project Proposal
Progress I Presentation	- Software project plan document - Software requirement document - Software design document - Software test plan document - Software test record document - Software traceability document
Progress II Presentation	- Software project plan document - Software requirement document - Software design document - Software test plan document - Software test record document - Software traceability document
Final Progress	- Final Progress Report - Software product - Project document

6.8.2 Milestone

Phase 1. Proposal: 1 March 2024 – 3 May 2024 = 64 days

Phase 2. Progress I: 4 May 2024 – 14 June 2024 = 42 days

Phase 3. Progress II: 1 July 2024 – 30 August 2024 = 61 days

Phase 4. Final Progress: 25 September 2024 – 29 October 2024 = 35 days

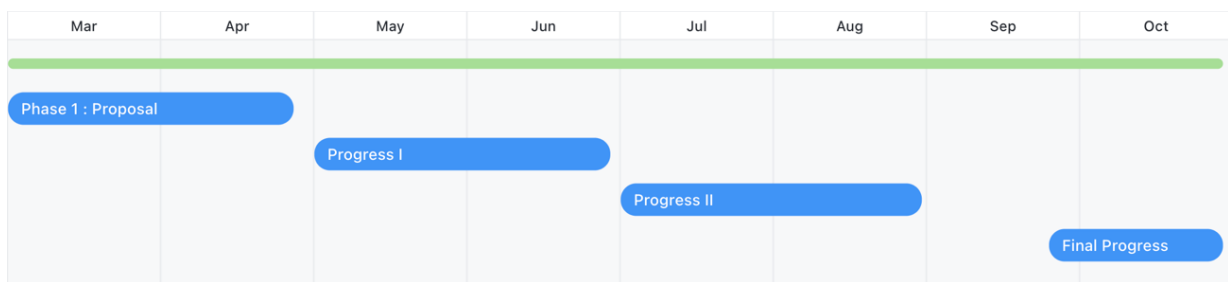


Figure 30 Milestone

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Phase 1. Proposal: 1 March 2024 – 3 May 2024 = 64 days

Tasks	Start date	End date	Duration	Responsibility
Research and writing proposal document	1 March 2024	5 April 2024	36	NS, WS
Prepare for the Presentation	6 April 2024	9 April 2024	4	NS, WS
Edit proposal document	10 April 2024	26 April 2024	17	NS, WS
Prepare for the Presentation proposal 2	27 April 2024	3 May 2024	7	NS, WS

Phase 2. Progress I: 4 May 2024 – 14 June 2024 = 42 days

Tasks	Start date	End date	Duration	Responsibility
Software requirement specification document	4 May 2024	7 May 2024	4	NS, WS
Project plan document	8 May 2024	10 May 2024	3	NS, WS
Software design document	11 May 2024	15 May 2024	5	NS, WS
Software test plan document	16 May 2024	19 May 2024	4	NS, WS
Feature #1 Dynamic Lyric Composition implement	20 May 2024	26 May 2024	7	NS, WS
Feature #1 Dynamic Lyric Composition test	27 May 2024	28 May 2024	2	NS, WS
Feature #2 Mood Detection Camera Integration implement	29 May 2024	5 June 2024	8	NS, WS
Feature #2 Mood Detection Camera Integration test	6 June 2024	7 June 2024	2	NS, WS

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Prepared Progress I report and presentation	8 June 2024	14 June 2024	7	NS, WS
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Phase 3. Progress II: 30 June 2024 – 30 August 2024 = 62 days

Tasks	Start date	End date	Duration	Responsibility
Software requirement specification document	30 June 2024	5 July 2024	6	NS, WS
Project plan document	6 July 2024	11 July 2024	6	NS, WS
Software design document	12 July 2024	16 July 2024	5	NS, WS
Software test plan document	17 July 2024	21 July 2024	5	NS, WS
Feature #3 Chord progression implement.	22 July 2024	2 August 2024	12	NS, WS
Feature #3 Chord progression test	3 August 2024	9 August 2024	7	NS, WS
Feature #4 AI Lyrics Analysis implement	10 August 2024	19 August 2024	10	NS, WS
Feature #4 AI Lyrics Analysis test	20 August 2024	25 August 2024	6	NS, WS
Prepared Progress II report and presentation	26 August 2024	30 August 2024	5	NS, WS

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Phase 4. Final Progress: 25 September 2024 – 29 October 2024 = 35 days

Tasks	Start date	End date	Duration	Responsibility
Wrap up and review all of the documents	25 September 2024	2 October 2024	8	NS, WS
Prepared final report and presentation	3 October 2024	29 October 2024	27	NS, WS

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